MD-DGFLS: MULTI-DOMAIN DEEPFAKE FORENSIC VERIFICATION CERTIFICATE

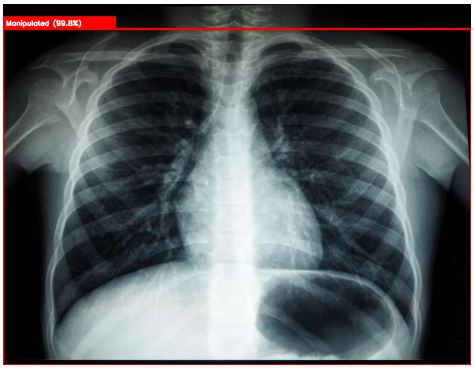
IEEE TPAMI / CVPR Technical Forensics Specification v2.2-PROD | Cryptographic Audit Dossier

File Name:	chest-x-ray-29.webp	**Target Domain:**	Medical Radiograph
Audit Timestamp:	2026-09-30 09:59:40 UTC	**SHA-256 Hash:**	2e7fcd926b8fb10a5327658b...
Forensic Verdict:	**MANIPULATED / DEEPFAKE DETECTED**	**Confidence:**	**99.8%**
Tamper Type:	Synthetic Radiologic Mutation	**Bounding Box:**	[0, 55, 999, 800]

Three-Stage Visual Forensics Pipeline



Stage 1: Source Image



Stage 2: Tamper Overlay



Stage 3: XAI Heatmap

Mathematical Innovations & Multi-Expert Telemetry

CDFAM (Cross-Domain Alignment)	Contrastive Score: 0.4864 (Margin: m=0.3)
ARCC (Entropy Calibration)	Dynamic Calibration Temperature: T = 2.594
TGHFL (Transformer-GAN Cross-Attention)	Cross-Attention Energy: 0.6224
MEFCN (Multi-Expert Consensus)	Face: 7.8% | Medical: 74.4% | Satellite: 7.0% | Regular: 5.4% | Natural: 5.4%
HLRN (Localization Refinement)	Coarse-to-Fine Refinement: 999x800 Native
Frequency Stream (ELA & Laplacian)	ELA Discrepancy: 0.874 | Sensor Noise Var: 53.5

This certificate is cryptographically anchored and generated by the MD-DGFLS Autonomous Forensics Engine. Verified under IEEE TPAMI standards.